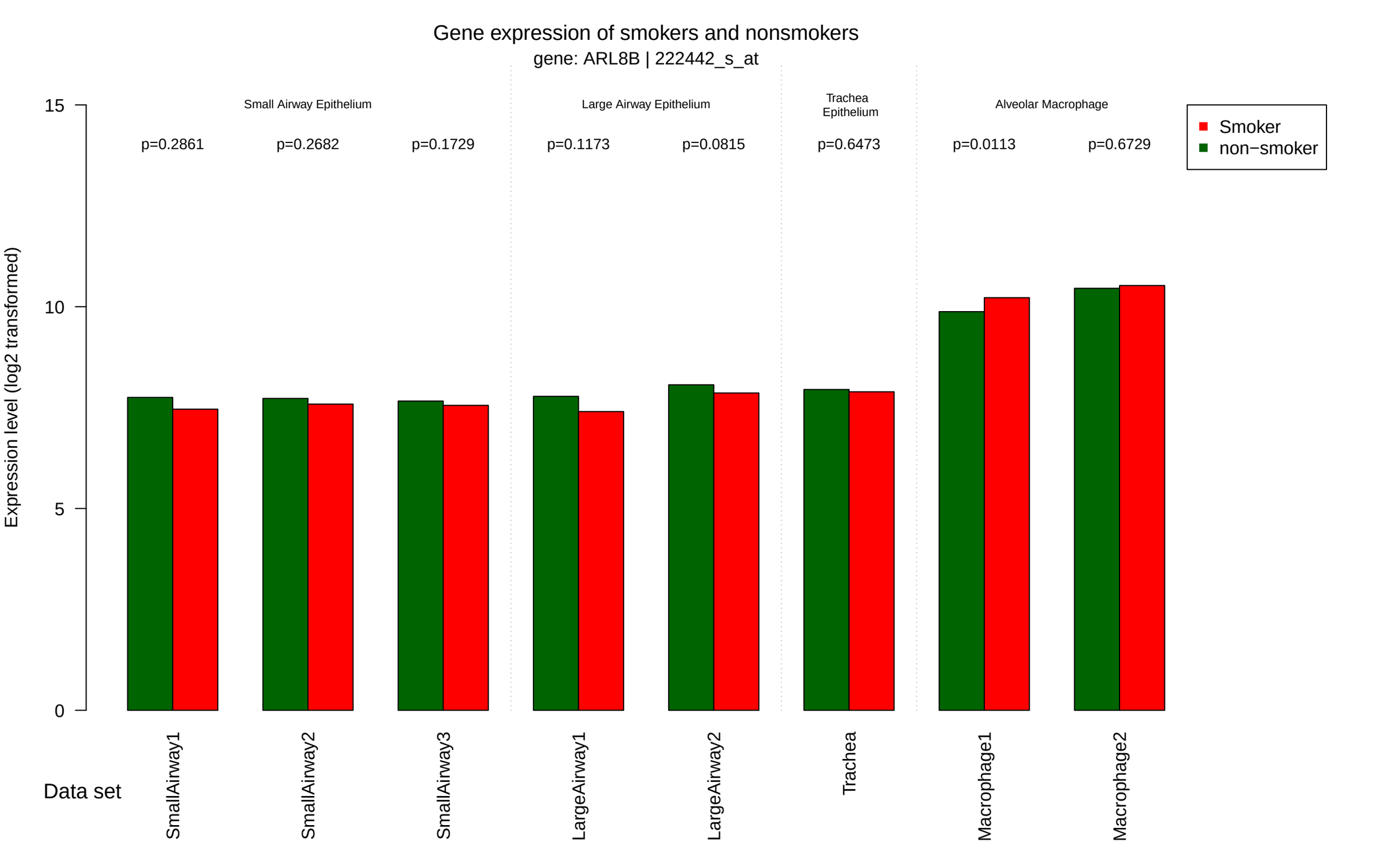
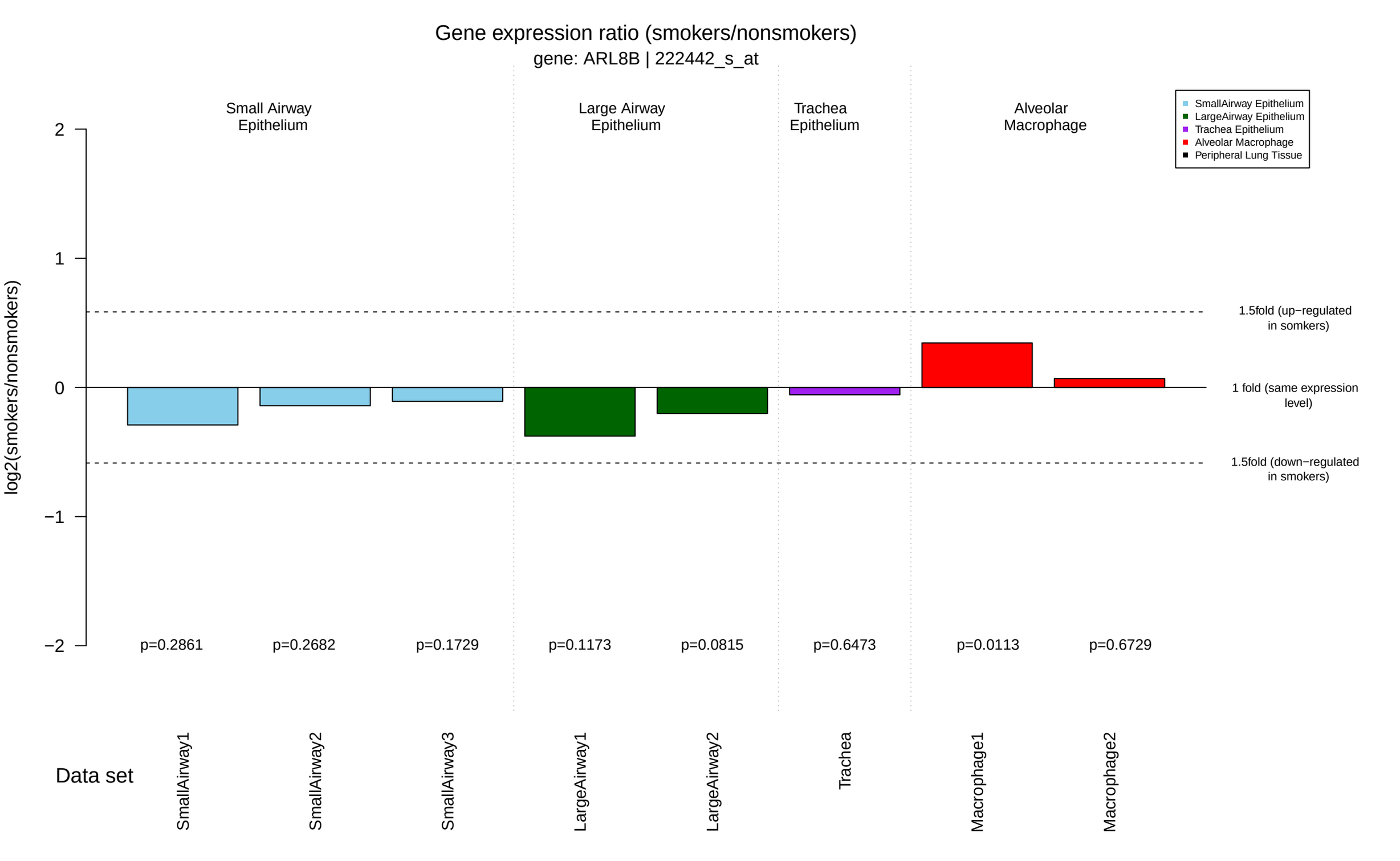
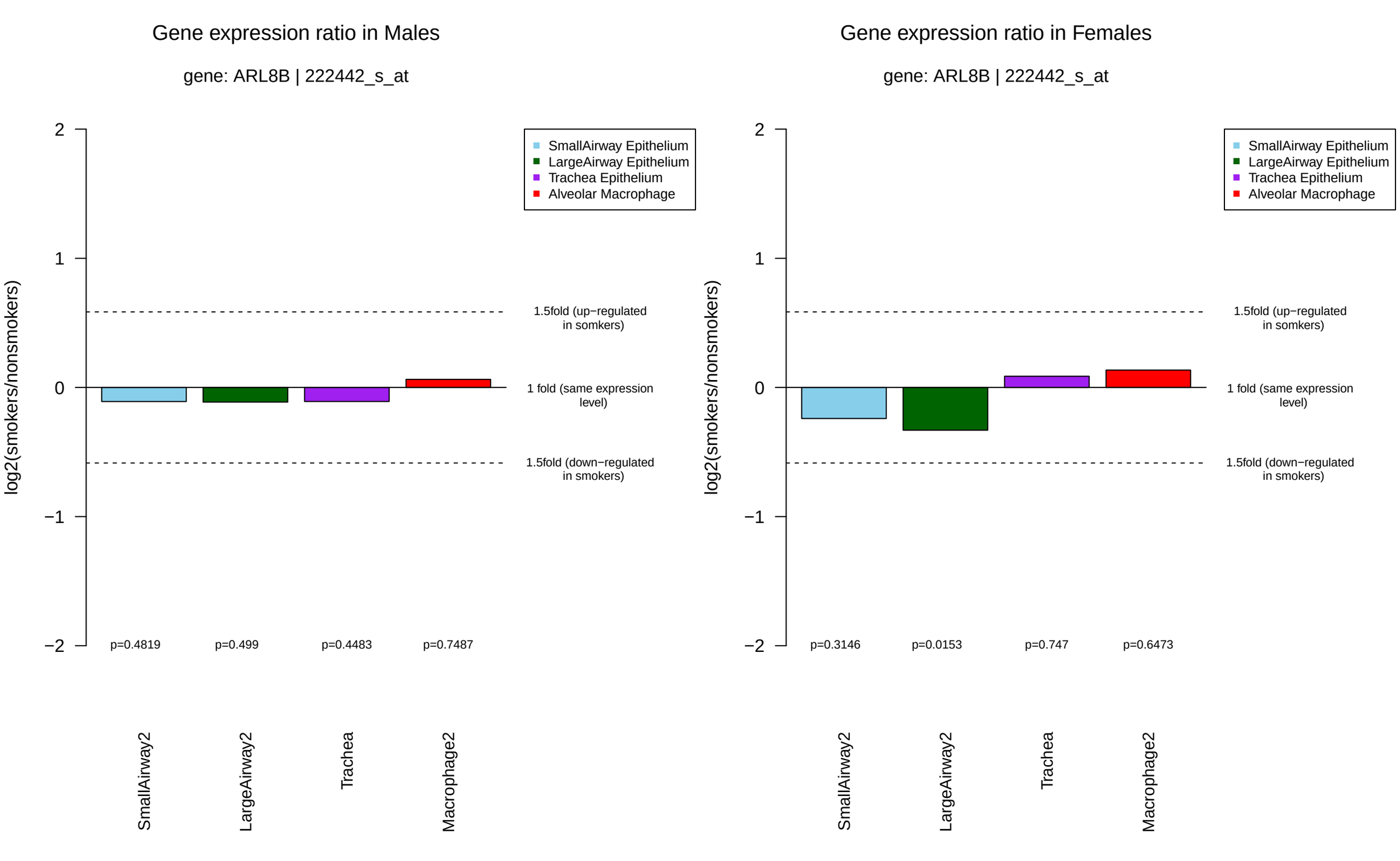
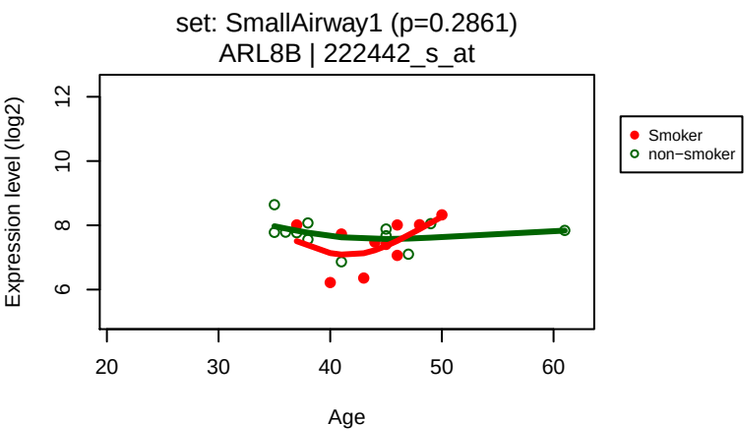
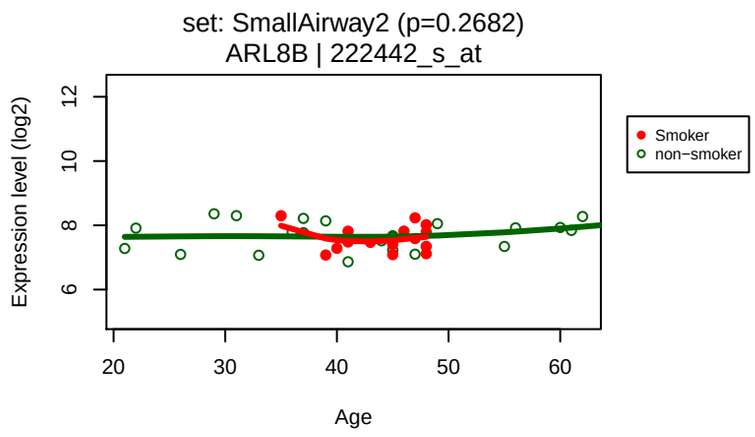
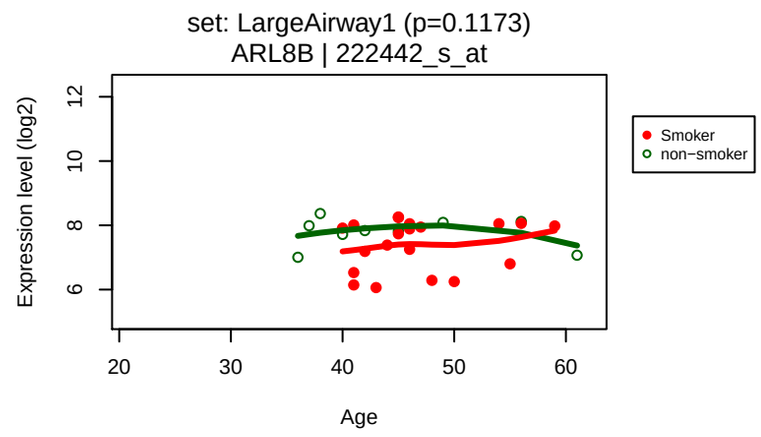

Gene expression of smokers and nonsmokers
gene: ARL8B | 222442_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2861
p=0.2682
p=0.1729
p=0.1173
p=0.0815
p=0.6473
p=0.0113
p=0.6729
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

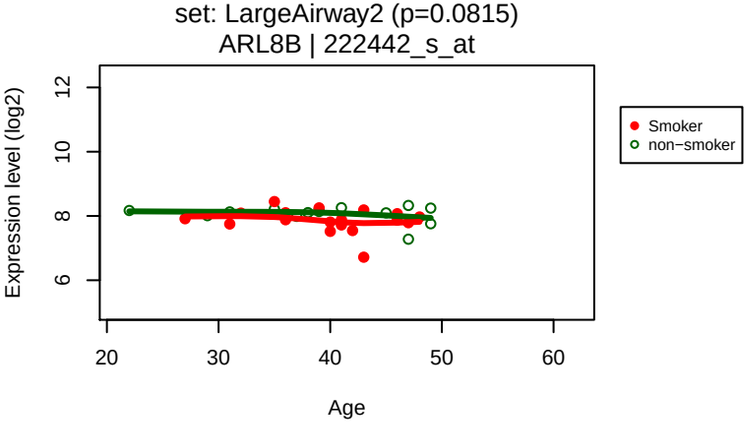

Gene expression ratio (smokers/nonsmokers)
gene: ARL8B | 222442_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2861
p=0.2682
p=0.1729
p=0.1173
p=0.0815
p=0.6473
p=0.0113
p=0.6729
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

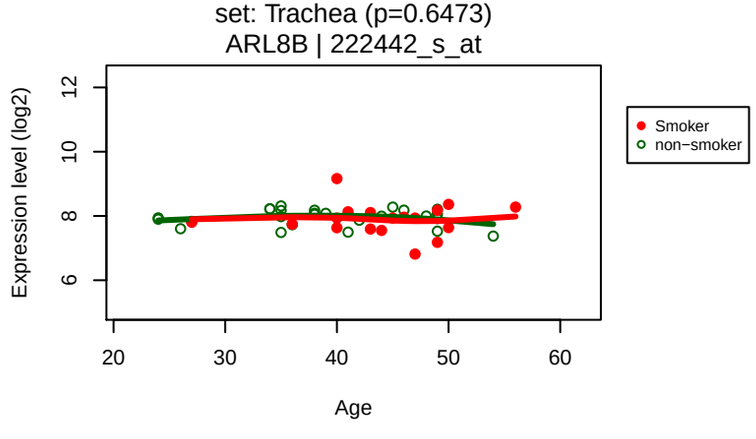

Gene expression ratio in Males
gene: ARL8B | 222442_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4819
p=0.499
p=0.4483
p=0.7487
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARL8B | 222442_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3146
p=0.0153
p=0.747
p=0.6473
SmallAirway2
LargeAirway2
Trachea
Macrophage2

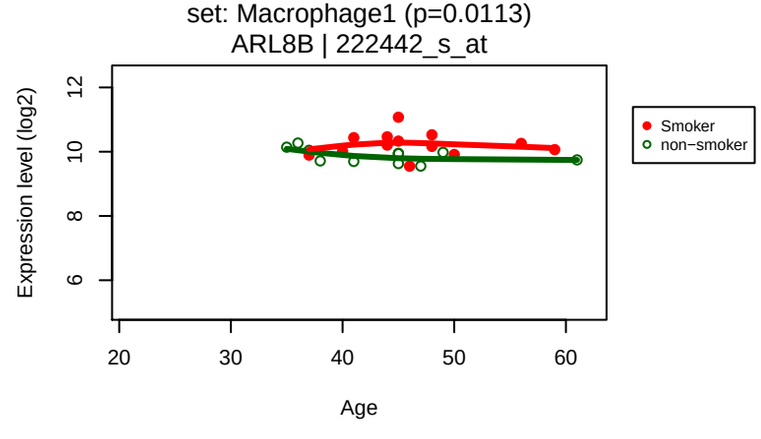

set: SmallAirway1 (p=0.2861)
ARL8B | 222442_s_at
Expression level (log2)
Age
Smoker
non−smoker

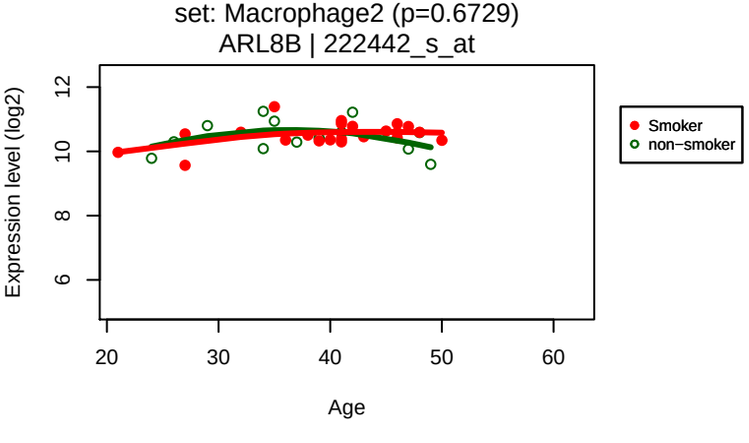

set: SmallAirway2 (p=0.2682)
ARL8B | 222442_s_at
Expression level (log2)
Age
Smoker
non−smoker

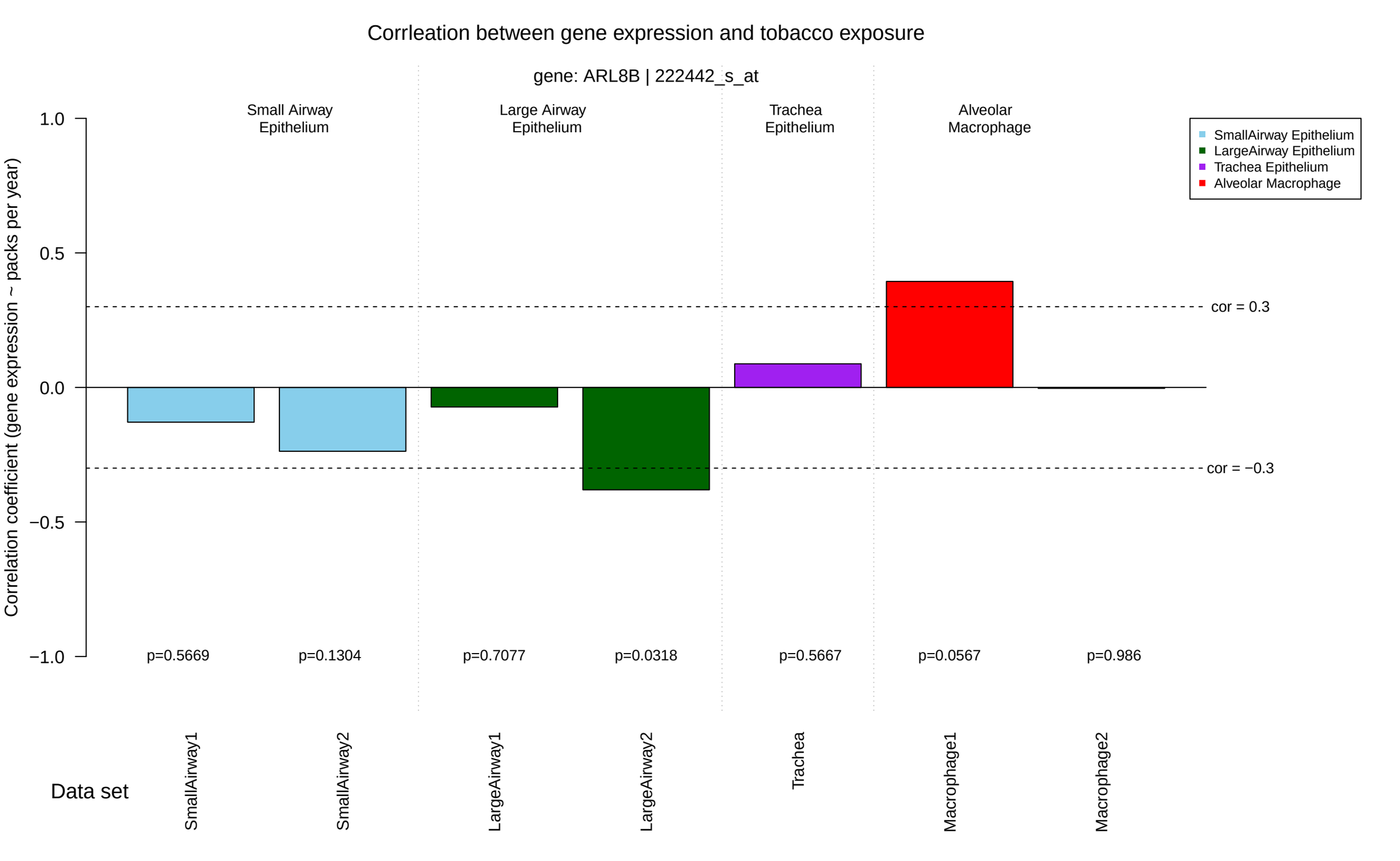

set: LargeAirway1 (p=0.1173)
ARL8B | 222442_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0815)
ARL8B | 222442_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6473)
ARL8B | 222442_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0113)
ARL8B | 222442_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6729)
ARL8B | 222442_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARL8B | 222442_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5669
p=0.1304
p=0.7077
p=0.0318
p=0.5667
p=0.0567
p=0.986
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2